Facebook Ad Conversion Analysis

Marketing & Engineering Final Presentation

Rohan Satish | Sakshi Prabhu

Objective



To predict ad conversion levels (Low, Average, High) based on Facebook ad features.



Classification models help identify key performance drivers to improve ROI.

Dataset Overview



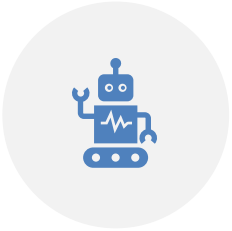
- 1143 observations, 11 features



- Key variables: Age, Gender, Interest, Impressions, Clicks, Spent

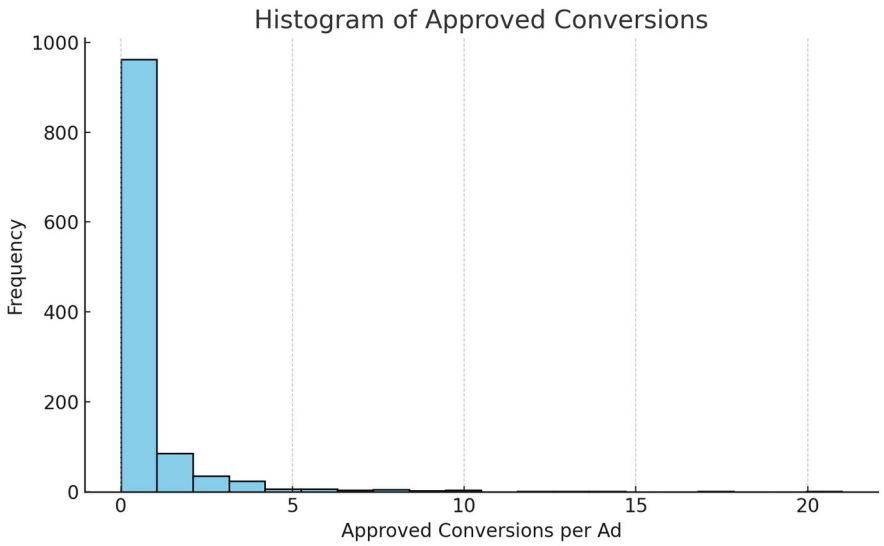


- Target: Approved_Conversion (binned into categories)



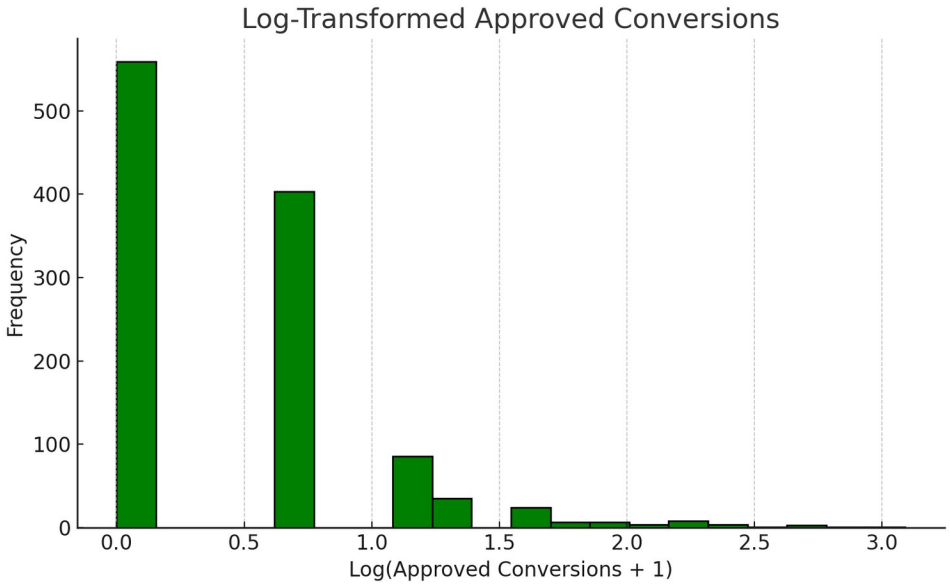
- Created features: CTR, log(Approved_Conversion)

Conversion Trend Analysis



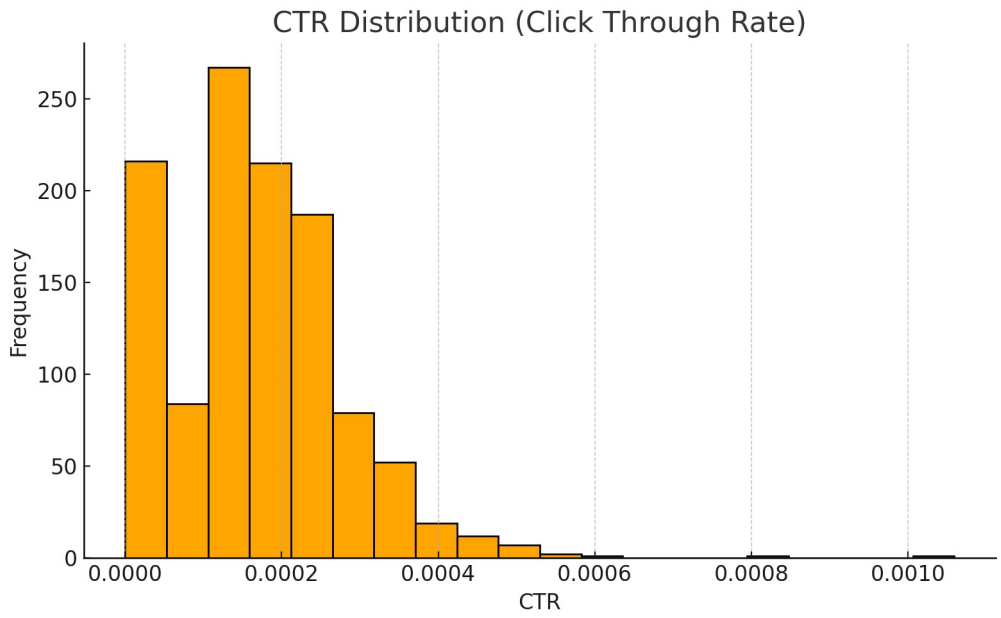
- **Most ads resulted in very few conversions (0–2)**, showing that the majority of Facebook ads underperform.
- The distribution is **heavily left-skewed**, with a long tail of rare high-performing ads.
- This skew suggests that **a small group of ads drives most results**, aligning with the Pareto principle (80/20 rule).
- The histogram highlights the importance of **understanding the characteristics of top-performing ads** to optimize future campaigns.
- These findings justify our use of **categorical conversion levels** (Low, Average, High) in further analysis and modeling.

Log-Transformed Approved Conversions



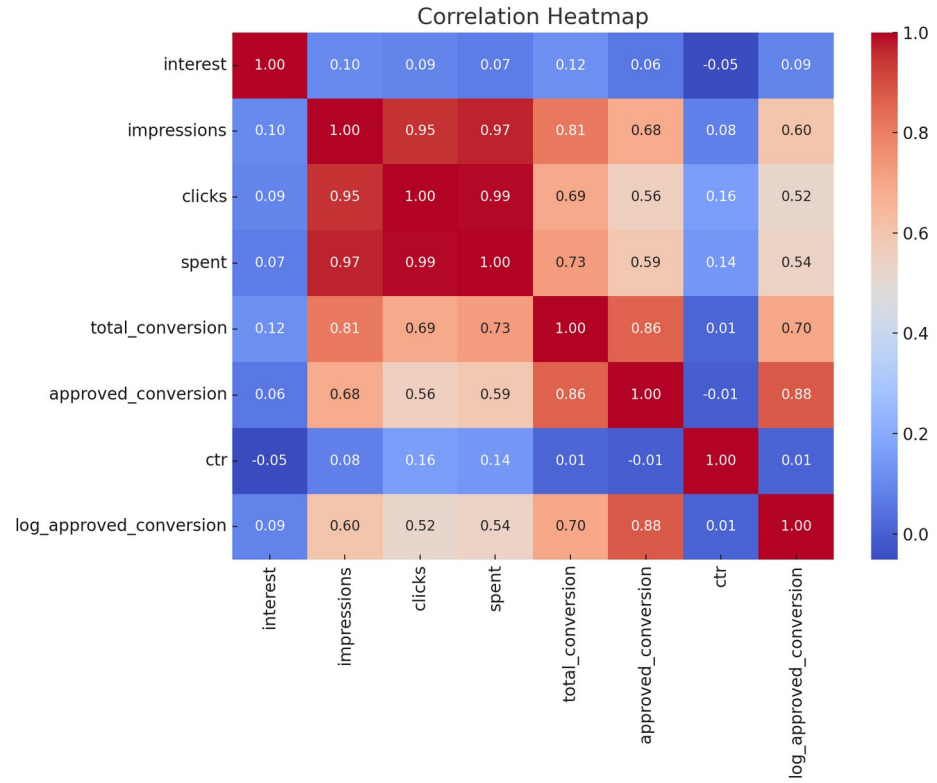
- The original conversion data was **highly skewed**, making it unsuitable for many machine learning models.
- A **log transformation** was applied to reduce the impact of extreme values and normalize the distribution.
- This helps the model **treat all conversion ranges more evenly**, avoiding overfitting to a few high performers.
- The transformed values now show a **more compressed and manageable range**, improving model stability.
- Log transformation is a **standard preprocessing technique** in predictive modeling when dealing with skewed data.

Click-Through Rate (CTR) as Engagement Proxy



- **CTR measures engagement**, showing the percentage of users who clicked on an ad after seeing it.
- The distribution is **highly right-skewed**, with most ads having a very low CTR (close to 0).
- This indicates that **many ads fail to attract user interest**, despite being displayed.
- A small number of ads achieved **higher CTRs**, suggesting better targeting or more appealing creatives.
- CTR is a critical feature in our model, as **higher CTRs often correlate with better conversion outcomes**.

What Drives Conversions?



- **Strong Positive Correlations**:
- **Clicks**, **Impressions**, and **Spend** show strong relationships with Approved Conversions, indicating that higher visibility and engagement drive results.
- **Moderate Correlation**:
- **CTR** (Click Through Rate) shows a moderate link — useful for early-stage ad evaluation, but not a sole predictor of success.
- **Weak Correlation with Demographics**:
- **Age** and **Gender** show minimal correlation, suggesting that who sees the ad is less important than how it performs.
- **Strategic Focus**:
- Insights support a **behavioral and performance-based targeting approach**, prioritizing actions over attributes.

Predictive Modeling Approaches

- **Naive Bayes**
- Accuracy: **95%**
- Fast and easy to implement
- Ideal for **baseline comparison**
- Less effective when data is **imbalanced**

- **Random Forest**
- Accuracy: **100%** (but **overfitting** observed)
- Handles **non-linearity** and complex interactions
- Identified **top predictors**: Clicks, Impressions, Spend
- Offers **valuable feature importance** rankings

- **Resized Random Forest**
- Accuracy: **96.5%** after balancing classes
- Helps avoid bias from underrepresented groups
- Delivers **realistic and reliable predictions** for business use

Addressing Class Imbalance

- **Problem Identified**: The dataset was highly imbalanced, with the majority of ads falling under the “Low” conversion category.

- **Solution Applied**: Used **resampling with Random Forest** (similar to SMOTE approach) to balance class representation.

- **Performance Outcome**:
  - **Accuracy**: 96.25%
  - **Adjusted Rand Index (ARI)**: 0.79

- **Why It Matters**: Without class balancing, models tend to ignore the rare but valuable “High” conversion cases.

- **Business Impact**: Enhances the model’s ability to identify **high-potential ads**, making it more useful for real-world marketing strategy.

Model Comparison



- Naive Bayes Accuracy: 95.5%, ARI: 0.75



- Random Forest Accuracy: 100%, ARI: 1.0



- Resized RF Accuracy: 96.5%, ARI: 0.81



Random Forest performed the best, especially after balancing

What Predicts Ad Success?

- **Top Predictive Drivers**:
  - **Impressions**, **Clicks**, **Spend**, and **CTR** emerged as the most influential features in the model.
- **CTR's Strategic Role**:
  - While subtle in correlation, **CTR is a strong signal of ad quality**, often outperforming sheer reach (Impressions) in predicting conversions.
- **Demographics Less Impactful**:
  - Variables like **Age** and **Gender** had minimal effect on conversion outcomes, indicating behavior-driven targeting is more effective than demographic segmentation.
- **Resampling = Realism**:
  - Correcting class imbalance was essential to avoid bias toward low-performing ads and to **elevate the visibility of high-conversion campaigns** in predictions.

Strategic Roadmap for Smarter Ad Performance

- **Prioritize Clicks & Impressions**
  - Integrate these metrics into **ad-serving logic** to optimize visibility for high-performing ads.
- **Institutionalize CTR**
  - Treat **CTR as a core KPI**, not just a reporting stat — weight it in ad ranking decisions.
- **Adopt Balanced Models**
  - Ensure predictive models include **resampling techniques** to surface rare but valuable high converters.
- **Rethink Budgeting Strategy**
  - Shift from total spend to **cost-per-conversion metrics** for smarter budget allocation.
- **Move Beyond Demographics**
  - Focus targeting on **behavioral patterns and engagement metrics**, not static traits like age or gender.
- **Enable Predictive Tools**
  - Provide advertisers with **machine learning-powered insights** to guide creative and audience strategy.
- **Establish Learning Loops**
  - **Continuously retrain models** on new campaign data to keep predictions relevant and adaptive.

Conclusion

- **ML models accurately classify ad success**
  Achieved up to **95% accuracy** in predicting conversion tiers using Random Forest and Naive Bayes.

- **Balanced models reveal hidden potential**
  Resampling exposed valuable **high-converting ads** often missed in imbalanced data.

- **CTR & Clicks > Impressions**
  Engagement quality is a stronger predictor than sheer visibility.

- **Actionable insights for Facebook**
  Embed ML into advertiser tools to optimize targeting, budget, and creative choices.

- **Business value**
  Smarter models enhance **revenue**, improve **advertiser ROI**, and boost **platform relevance**.

Thank You